1 Data Generation

1.1 Pattern Matching

A *pattern* is a set.

1.1.1 Example

Let $a \circledast b = [a_1, \cdots, a_n, b_1, \cdots, b_m]$, where $a = [a_1, \cdots, a_n]$ and $b = [b_1, \cdots, b_m]$.

$$
\begin{array}{rcl}
\mathfrak{E} & := & \mathfrak{T} \cup \{\mathfrak{e}_1 \circledast \mathfrak{o} \circledast \mathfrak{e}_2 \,|\, \mathfrak{o} \in \{[\text{“}+\text{”}], [\text{“}-\text{”}]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\
\mathfrak{T} & := & \mathfrak{F} \cup \{\mathfrak{t}_1 \circledast \mathfrak{o} \circledast \mathfrak{t}_2 \,|\, \mathfrak{o} \in \{[\text{“}\times\text{”}], [\text{“}\div\text{”}]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\
\mathfrak{F} & := & \mathfrak{N} \cup \{[\text{“(”}] \circledast \mathfrak{e} \circledast [\text{“)”}] \,|\, \mathfrak{e} \in \mathfrak{E}\} \\
\mathfrak{N} & := & N \cup N^* \\
N & := & \{\text{“0”}, \cdots, \text{“9”}\}
\end{array}
$$

What is the difference between pattern matching and tokenwize parsing?

2 gen.jl and pattern.jl

A *grimoire* of a pattern $\mathfrak{P}$ is a function $\Phi : L_\Phi \to \mathcal{P}(\mathfrak{P})$, where $L_\Phi \subseteq \mathcal{P}(\mathfrak{P})$ is a set of *labels*[1].

Let $G := (\langle \mathscr{G} \rangle, *)$ be a semigroup. Let $\triangle : \boxed{\Phi} \times G^2 \to \boxed{\Phi}$ – where $\boxed{\Phi}$ is the set of all grimoires of a pattern $\mathfrak{P} \subseteq G^{<\omega}$!!!!!!should G be $\mathscr{G}$????, in either case, $\mathfrak{P}$ should be defined better!!!!!– be defined for some grimoire Φ by $\forall l \in L_\Phi \setminus \{l_1 * l_2\}\ \ (\Phi'(l) = \Phi(l))$ and $\Phi'(l_1 * l_2) = \Phi(l_1 * l_2) \cup \{[l_1, l_2]\}$ if $l_1 * l_2 \in L_\Phi$ and $\Phi'(l_1 * l_2) = \{[l_1, l_2]\}$ if not – where $\Phi' = \triangle(\Phi, [l_1, l_2])$ and $l_1, l_2 \in G$. Note that since $L_\Phi = \mathrm{dom}(\Phi)$, then $L_{\Phi'} = L_\Phi \cup \{l_1 * l_2\}$ if $\Phi' = \triangle(\Phi, [l_1, l_2])$.

?????$\Phi : \mathscr{G} \to \{\emptyset\}$?????Now, consider the exploration function $\mathscr{E}$: $\boxed{\Phi} \times \boxed{\mathcal{C}}^{<\omega} \to \boxed{\Phi}$ defined by $\mathscr{E}(\Phi, [\mathcal{C}_1, \cdots, \mathcal{C}_n]) = \bigcirc_{i=1}^{n} \triangle\left(\Phi, \mathcal{C}_i\left(L_\Phi\right)\right)$!!!!!!This is another subgroup!!!, where $\boxed{\mathcal{C}} \subset \mathcal{P}(G) \to G^2$ and $\forall \mathcal{C} \in \boxed{\mathcal{C}}\ \Big(\forall S \in \mathcal{P}(G)\ \big(\mathcal{C}(S) \in S\big)\Big)$. Then !!!!!THIS SHOULD BE A LEMMA!!!Φ_n is a partial or full representation of the group G, where each element in $L(\Phi_n)$ maps to the set of ordered pairs of elements in G to which it is mapped by $*$.

!!!!!!!!! therefore, for each output generated by $\mathscr{E}$, one can define a function $\lightning$ such that: !!!!!!!!Let a function $\lightning : \boxed{\Phi} \times G^{<\omega} \times \boxed{\Pi} \to \mathscr{G}^{<\omega}$ be defined by

$$\lightning(\Phi, \sigma, \Pi) = \begin{cases} \sigma & : \ \forall l \in \sigma\ \ (l \in \mathscr{G}) \\ \lightning\left(\Phi, \lightning_{\text{step}}\left(\Phi, \sigma, \Pi\right), \Pi\right) & : \ \text{otherwise} \end{cases},$$

where $\lightning_{\text{step}}\left(\Phi, \sigma, \Pi\right) = \bigcircledast_{i=1}^{|\sigma|} \Pi\left(\Phi\left(\sigma_i\right)\right)$ and $\boxed{\Pi} \subset \mathcal{P}\left(G^{<\omega}\right) \to G^{<\omega}$

[1]Lables – or, *concepts* – are useful abstractions of patterns, which will be used later to optimize the complexity of solution-finding systems. The subsets $\mathfrak{T}, \mathfrak{F}, \mathfrak{N}$ and N in example 1.1.1 are examples of lables.

where $\forall \Pi \in \boxed{\Pi} \left(\forall S \in \mathcal{P}\left(G^{<\omega}\right) \ \left(\Pi(S) \in S\right)\right)$.

!!!!!BEVISA FAKTUM OM DEN GENERERADE STRUKTUREN.

SKRIV EXEMPEL FÖR ATT DEMONSTRERA HUR EN GRIMOIRE BYGGS UPP AV $\mathscr{E}$, OCH SEDAN HUR ϟ GENERERAR EN LÖSNING – T.EX. FÖR EN PERMUTATIONSGRUPP (ELLER NÅGON ANNAN SYNTAX).!!!!!!